Report Milestone – 4
Analytics Output & Dashboard Visualization Using Power BI

TEAM 3

Mohesh V K - 2362628
Praneeth – 2362634
Ramalingam K - 2362617
Jency A - 2361016

1. Overview

This milestone focuses on the development of an interactive data visualization system using Microsoft Power BI to analyse environmental sensor data. The objective is to transform raw data into meaningful insights by designing multiple dashboards that represent different environmental risk conditions—Safe, Moderate, and High—along with a consolidated analytical view.

The dashboards are designed to support interpretability, enable pattern recognition, and provide a clear understanding of how environmental parameters influence risk levels.

2. Dataset Description and Preparation

The dataset consists of time-series environmental readings with the following attributes:

- Timestamp
- Temperature
- Humidity
- Air Quality (ppm)
- Pressure (hPa)
- Prediction (Safe / Moderate / High)

Data preparation involved ensuring appropriate data types for all attributes, particularly maintaining the timestamp in a valid date-time format and numerical consistency for all sensor readings. Aggregation behaviour was carefully controlled to avoid distortion in visual outputs, ensuring that trends and distributions reflect actual variations in the dataset.

3. Dashboard Architecture

The Power BI report is structured into four distinct dashboards:

1. Safe Condition Dashboard
2. Moderate Condition Dashboard
3. High Risk Dashboard
4. Combined (Common) Dashboard

Each dashboard isolates or integrates risk categories to provide both focused and holistic analytical perspectives.

4. Safe Condition Dashboard

Purpose

The Safe dashboard represents environmental conditions where all parameters remain within acceptable thresholds, indicating no immediate risk.

Design and Visual Components

The dashboard is filtered to include only records classified as "Safe." The following visuals are used:

- **KPI Indicators** present average values of temperature, humidity, air quality, and pressure. These indicators provide a concise summary of stable environmental conditions.
- **Time-Series Line Chart** illustrates the progression of temperature, humidity, and air quality over time. The absence of sharp fluctuations reflects environmental stability.
- **Gauge Visualization** for air quality confirms that values consistently fall within the safe range, reinforcing the classification.
- **Scatter Plot** (Temperature vs. Air Quality) demonstrates a non-critical relationship between variables, with all points confined to a safe region.
- **Distribution Chart** shows that all observations belong to the safe category, validating the applied filtering.

Interpretation

This dashboard confirms that environmental conditions are stable, predictable, and within safe operational limits. It serves as a baseline reference for comparing more critical scenarios.

5. Moderate Condition Dashboard

Purpose

The Moderate dashboard captures transitional environmental states where parameters begin to deviate from optimal ranges but do not yet pose severe risk.

Design and Visual Components

Filtered to “Moderate” predictions, this dashboard includes:

- **KPI Indicators** reflecting elevated but non-critical parameter values.
- **Line Chart** showing noticeable fluctuations and gradual upward trends, indicating increasing variability.
- **Gauge Visualization** positioning air quality within intermediate thresholds.
- **Scatter Plot** revealing clustering patterns that suggest emerging relationships between temperature and air quality.
- **Distribution Visualization** confirming the dataset is entirely within the moderate category.

Interpretation

This dashboard highlights early warning conditions. While not immediately dangerous, the observed trends suggest that continued deviation could lead to high-risk states. It emphasizes the importance of monitoring and preventive action.

6. High Risk Dashboard

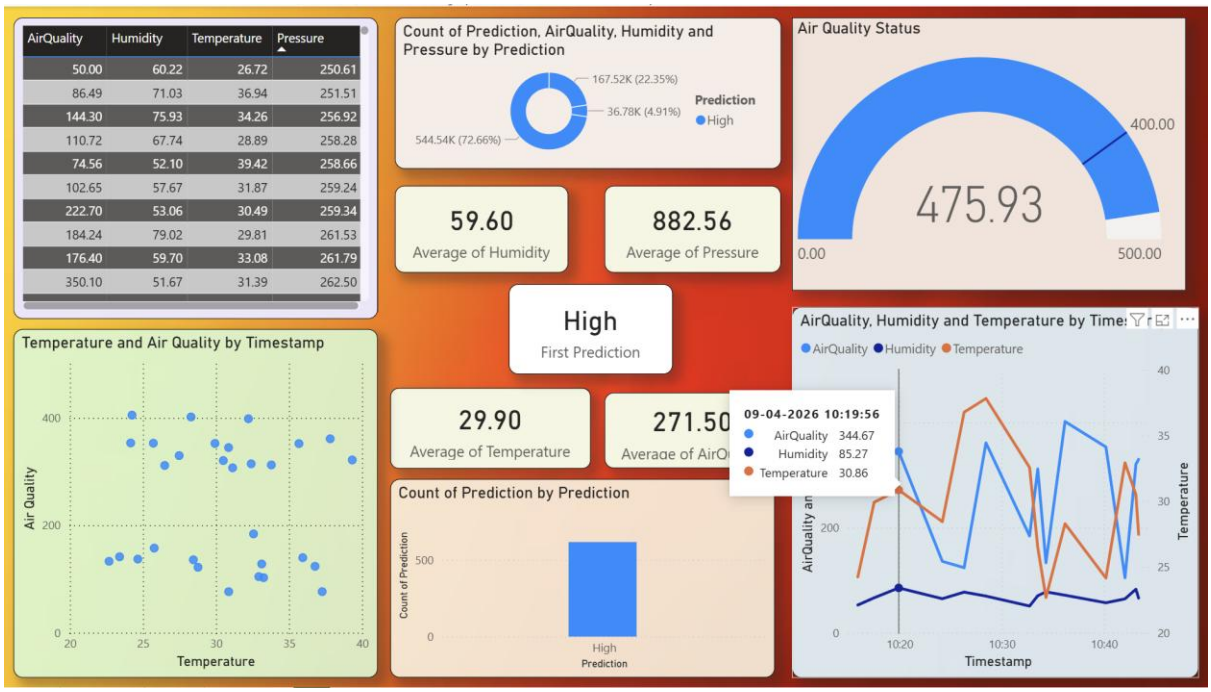
Purpose

The High Risk dashboard focuses on critical environmental conditions where one or more parameters exceed safe thresholds.

Design and Visual Components

With filtering applied to “High” predictions, the dashboard includes:

- **KPI Indicators** showing significantly elevated values, particularly in air quality and temperature.
- **Line Chart** characterized by sharp spikes and irregular patterns, indicating instability.
- **Gauge Visualization** placing air quality firmly within hazardous ranges.
- **Scatter Plot** displaying strong correlations between variables that contribute to risk escalation.
- **Distribution Visualization** confirming that all records fall within the high-risk category.



Interpretation

This dashboard provides clear evidence of unsafe environmental conditions. The visual patterns indicate rapid changes and extreme values, supporting the classification of high risk and the need for immediate intervention.

7. Combined (Common) Dashboard

Purpose

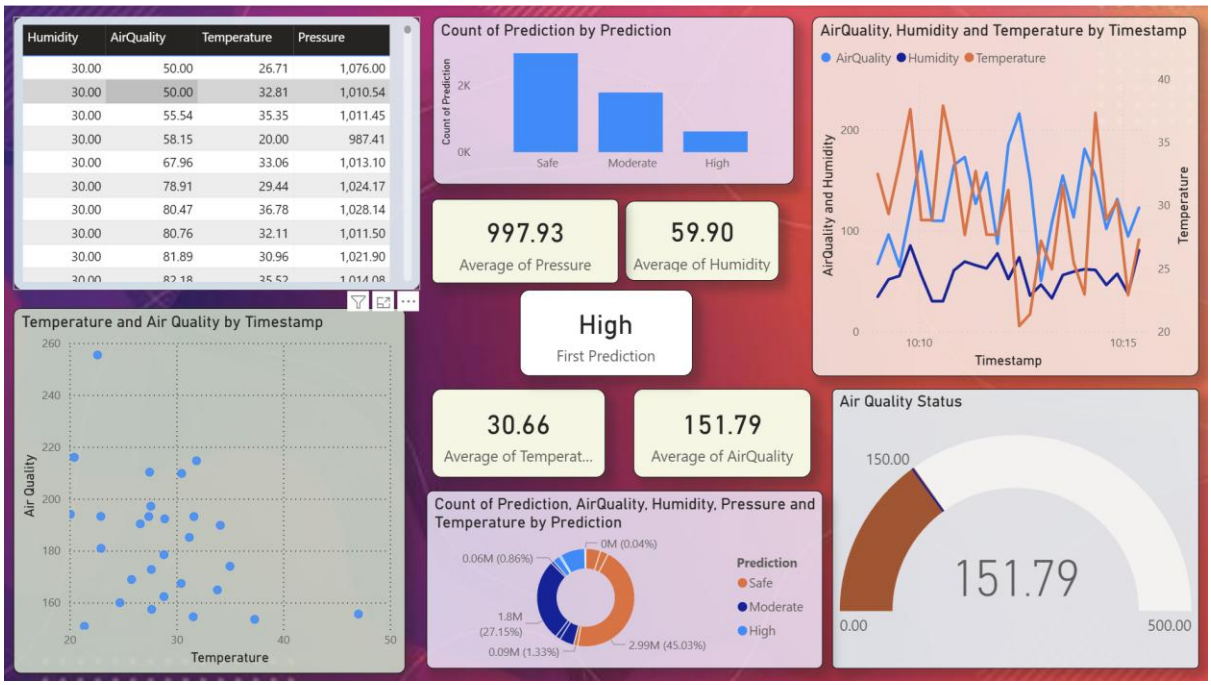
The Combined dashboard integrates all risk categories to provide a comprehensive view of environmental behaviour across different conditions.

Design and Visual Components

This dashboard is not filtered, allowing full dataset analysis:

- **KPI Indicators** summarize overall system performance across all conditions.
- **Multi-Series Line Chart** (Air Quality over time with risk-based segmentation) highlights transitions between safe, moderate, and high states.
- **Donut Chart** represents the proportional distribution of risk categories, offering a quick overview of system behaviour.
- **Scatter Plot** (Temperature vs. Air Quality with categorical colouring) reveals how different regions correspond to varying risk levels.

- **Gauge Visualization** provides a real-time representation of air quality status relative to defined thresholds.
- **Stacked Column Chart** compares average parameter values across risk categories, emphasizing differences in environmental conditions.
- **Tabular View** presents detailed records for validation and deeper inspection.



Interpretation

The combined dashboard enables holistic analysis by linking parameter trends with risk classification. It effectively demonstrates how variations in environmental factors lead to transitions between safety levels, making it the most comprehensive analytical view in the report.

8. Key Features and Analytical Capabilities

- Multi-level dashboard design for focused and comparative analysis
- Integration of categorical and continuous visualizations
- Clear mapping between environmental parameters and risk levels
- Interactive filtering enabling dynamic exploration
- Consistent visual encoding for intuitive interpretation

9. Conclusion

The Power BI implementation successfully converts raw environmental data into structured, interpretable insights. By segmenting the analysis into distinct risk-based dashboards and integrating them into a unified view, the system provides both depth and breadth in understanding environmental conditions.

The design emphasizes clarity, interactivity, and analytical relevance, making it suitable for monitoring applications and decision-support systems.